



Lab 03: Expressions

CSE 4108

Structured Programming I Lab



Lab Tasks

1. Reversed Four Ever:

Write a C program that uses takes input a four digit integer from the user and then prints the integer reversed.

Sample Run:

Enter 4 digit integer: 1234

Reversed 4 digit integer: 4321

***Hint:** If n is an integer, then $n\%10$ is the last digit in n and $n/10$ is n with the last digit removed.*

2. Secret Number:

Secret Agent Latent Talent (SALT) agency has started a new reality show as a platform to flourish upcoming secret agents. Every participating agent is assigned a secret number, which can be decoded to a single digit CODE. The secret number is of the format N ABCD WXYZ.

The formula to calculate CODE from secret number is to add A with Z, B with X, C with Y and D with W and then, add all these four sums

together again. Then one needs to take the remainder of the final sum with respect to N.

Sample Run:

Enter Secret Number: 9 1234 5678

CODE: 0

***Hint:** If a number 123456 is scanned with `scanf("%3d%2d%d", &a, &b, &c)`, then a will be 123, b will be 45 and c will be 6. In other words, in `scanf`, %nd scans the first n digits and leaves the next digits for the next specification(s).*

3. Hexadecimal conversion:

Write a program that reads an integer entered by the user and displays it in **hexadecimal** (base 16).

Sample run:

Enter a number between 0 and 32767: 1953

In hexadecimal, your number is: 007A1

The output should be displayed using 5 digits, even if fewer digits are sufficient.

Hint: To convert the number to **hexadecimal**, first divide it by 16; the remainder is the last digits of the hexadecimal number. Then divide the original number by 16 and repeat the process to arrive at the next-to-last digit.

(**printf** is capable of displaying numbers in base 16, as we'll see later in chapter 7 that there's actually an easier way to write this program.)

4. Dot Dot Dot:

Write a C program that would take two 3-Dimensional vectors and print the angle between them in degrees.

Sample run:

```
Enter Vector 1: 1.1, 2.2, 3.3  
Enter Vector 2: 3.3, 2.2, 1.1  
Angle (degree): 44.415318
```

5. The Swapperoo:

Write a program that stores two integers in two variables and then swaps the values of the two variables.

Hint: Use a temporary variable to store one value (a) and then overwrite a with other value (b). Assign the temporary variable to b.

6. The Swapperoo II (Temp not allowed):

Modify the program from the previous task so that it doesn't use any temporary variable.

***Hint:** Think of the problem from a mathematical perspective. A combination of addition/subtraction or multiplication/division should suffice*